

Yasien Ghalwash

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EDUCATION

- **Queensland University of Technology** Brisbane, Australia
Ph.D in Information Technology Sep 2024 – Present
 - **Thesis:** Rethinking Energy Harvesters as Sensors: A New Paradigm for Continuous Motion Tracking
 - **Supervisors:** Sara Khalifa and Raja Jurdak
- **Cairo University, Faculty of Engineering** Cairo, Egypt
Bachelor of Biomedical Engineering, GPA: 3.5/4.0 Sep 2018 – Jul 2023
 - **Courses:** Machine Learning, Computer Vision, Computer Graphics, Probability, Statistics, Calculus, Linear Algebra, Digital Signal Processing, Data Structures, Algorithms, Databases, Embedded Systems

RESEARCH AND INDUSTRY EXPERIENCE

- **Hamburg University of Technology (TUHH)** Australia & Germany
Research Collaborator, UA-DAAD Joint Research Project Jan 2026 – Present
 - Contributing to the UA-DAAD funded Australia-Germany joint research project: *Intermittently Powered & Configurable Machine Learning for Water Monitoring*.
 - Developing intermittently powered machine learning systems for energy-constrained environmental sensing platforms.
 - Collaborating with interdisciplinary researchers from QUT and TUHH on system design, implementation, and experimental evaluation.
 - Investigating adaptive and energy-aware ML algorithms for continuous sensing using energy harvesting devices.
- **Queensland University of Technology** Brisbane, Australia
Sessional Academic (Applied Internet of Things - IAB330) Jul 2025 – Nov 2025
 - Facilitated weekly practical sessions for undergraduate students, focusing on the design, deployment, and hardware-software integration of IoT systems.
 - Guided students in sensor interfacing, cloud connectivity, and data visualization using edge computing platforms.
 - Provided technical mentorship on student project prototypes, troubleshooting networking and hardware constraints.
- **Inspire for Solutions Development** Cairo, Egypt
AI Engineer Nov 2023 – Aug 2024
 - Fine-tuned Large Language Models (LLMs) using advanced NLP techniques to translate Saudi dialect Arabic into Modern Standard Arabic with high accuracy.
 - Implemented model compression techniques, including quantization and pruning, improving inference efficiency for resource-constrained deployments.
- **Astute Imaging** Remote
R&D Engineer Jul 2022 – Aug 2023
 - Designed and developed a CAD system architecture integrated with PACS to automate AI-based clinical decision workflows using Orthanc and FastAPI.
 - Optimized deep learning models for CPU/GPU deployment, reducing inference time from 3 minutes to 30 seconds (6× speedup).
 - Containerized system components using Docker and managed deployment on Microsoft Azure Virtual Machines.
 - Developed medical imaging models using U-Net, Attention U-Net, and EfficientNet architectures.
 - **Tools:** PyTorch, TensorFlow, FastAPI, OpenCV, Orthanc, Docker, Microsoft Azure

PROJECTS

- **Computer-Aided Breast Cancer Lesion Segmentation & Classification** (PACS, DICOM, FastAPI, Docker, TensorFlow, Azure) End-to-end PACS-integrated system for automatic segmentation and classification of breast cancer ultrasound images. Integrated with a DICOM viewer providing image processing tools for physicians. [🔗](#) (2023)
- **Face Recognition using Variational AutoEncoders** (TensorFlow, VAE) Facial detection model that learns latent variables in face datasets to adaptively re-sample training data for improved performance. [🔗](#) (2022)
- **LSTM Music Generator** (TensorFlow, RNN, LSTM) Trained an LSTM model to predict the next character in song sequences and generate new music compositions. [🔗](#) (2022)
- **GI Tract Image Segmentation** (TensorFlow, U-Net) Developed a segmentation model for MRI scans to isolate the stomach and intestines, enabling more precise radiation therapy and protecting healthy tissues. [🔗](#)
- **Autonomous Driving Car Detection** (TensorFlow, YOLO) Implemented the YOLO algorithm to extract bounding boxes and identify the real-time positions of vehicles for autonomous navigation systems. [🔗](#)

PUBLICATIONS

- **Ghalwash, Y.**, Khamis, A., Sandhu, M., Khalifa, S., Jurdak, R., & S. K. Das. “From power to position: Continuous trajectory tracking with wearable solar harvesters.” IEEE Transactions on Mobile Computing, under review 2026
[D1 Journal]
- **Ghalwash, Y.**, Khamis, A., Sandhu, M., Khalifa, S., & Jurdak, R. “SolarTrack: Exploring the Continuous Tracking Capabilities of Wearable Solar Harvesters.” In Proc. IEEE PerCom. 2026
[A* Conference] 
- **Ghalwash, Y.** “Rethinking Energy Harvesters as Sensors: A New Paradigm for Continuous Motion Tracking.” In Proc. IEEE PerCom PhD Forum. 2026
[A* Conference] 
- Salem, S., Mostafa, A., **Ghalwash, Y.**, et al. “Computer-Aided System for Breast Cancer Lesion Segmentation and Classification Using Ultrasound Images.” In Proc. IEEE EHB. 2023


HONORS AND AWARDS

- Student Travel Grant, IEEE PERCOM 2026
- Best Poster Award, QUT Energy Transition Centre Retreat 2025
- Best Presentation Award, IS Doctoral Consortium, School of Information Systems, QUT 2025
- HDR Accomplishment Award, School of Information Systems, QUT 2025

PRESENTATIONS

- IEEE PERCOM (Main Track) 2026
- IEEE PERCOM (PhD Forum) 2026
- QUT Energy Transition Centre Retreat 2025
- IS Doctoral Consortium, School of Information Systems, QUT 2025

LEADERSHIP AND COMMUNITY SERVICE

- **QUT Energy Transition Centre (ETC)** Brisbane, Australia
HDR Representative 2026 – Present
 - Represent Higher Degree Research (HDR) students within the centre, contributing to governance, feedback, and community-building initiatives.
 - Facilitate communication between HDR students and centre leadership, supporting research culture and collaboration.
- **Resala Charity Organization** Cairo, Egypt
Volunteer Team Lead Jul 2016 – Aug 2022
 - Led a team of 10 volunteer in the building and renovation of mosques and schools across Cairo over six years of sustained community service.
 - Coordinated fundraising campaigns and managed public-relations activities supporting the organisation’s educational and infrastructure programs.
- **STP (Science, Technology, Projects)** Cairo, Egypt
Internet of Things Instructor Sep 2020 – Oct 2021
 - Designed the IoT curriculum and practical workshop content in coordination with the instructional team.
 - Mentored participant teams through 10+ hands-on embedded systems and IoT projects.
 - Designed and led the capstone design competition in which participants built a smart parking system.

SKILLS SUMMARY

- Languages Python, C/C++, Java, SQL
- Frameworks & Libraries FastAPI, PyQt, Vtk.js, Scikit-learn, Pandas, PyTorch, TensorFlow, OpenCV, MONAI, WandB, Django, Flask
- Software Development OOP, Design Patterns, Git, Docker
- IoT & Embedded Systems STM32, ESP8266, Arduino, Raspberry Pi